MALANHACK 2026 · FINTECH · PROBLEM DEFINITION & SOLUTION FRAMEWORK

LienLine

When a cyber-fraud complaint names a bank account, freeze **only the money the trail proves**, not the whole account.

Asish Kumar · Akash Kumar · October 2026 · Live demo: lienline-desk.vercel.app

MP HIGH COURT · 27 JUL 2026 [1]

A liquor shop's current account

Frozen by the bank
₹2,51,00,000 → Disputed amount **₹980**

LienLine: lien ₹980, release the rest

WHERE THE BRIEF'S FIVE REQUIRED ITEMS ARE

1. **slides 2–4** — Problem statement definition
2. **slides 5–7** — Proposed technological architecture
3. **slide 9** — Global benchmarking, 6 countries
4. **slides 10–12** — Market size, revenue, costs
5. **slide 15** — Validated references, 25 sources

PROBLEM STATEMENT

India freezes fraud money fast, and freezes innocent money with it.

MARKED AS LIEN

₹8,189 cr

held on fraud complaints, 2021–25, out of ₹55,050 crore reported lost [4]

ONE SCAM

36,000+

accounts frozen after a single ₹4.43 crore fraud, small merchants included [6]

ONE HIGH COURT

105

freeze petitions decided together, with 14 binding guidelines [2]

GIVEN BACK

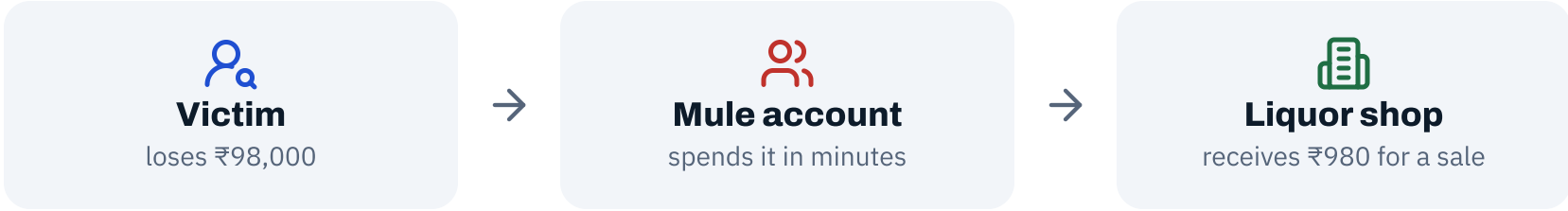
₹18 cr

restored in 36,290 cases, against ₹11,158 crore "saved" since 2021 [3]

Problem statement. Police notices tell banks to "freeze", often without an amount. Banks freeze whole accounts because no one computes how much of the fraud money each account actually holds.

PROBLEM STATEMENT · ROOT CAUSE

The money trail is never measured, so the safe choice for a bank is to freeze everything.



TODAY

Notice reaches the shop's bank via CFCFRMS. Bank cannot see the trail, so it freezes the account: **₹2.51 crore** locked for months [1]. Courts: "innocent recipients bear the brunt" [7].

WITH LIENLINE

Trace the trail across banks, compute what is traceable to the fraud in each account, **lien ₹980**, release the rest, start the grievance clocks [9], and document why.

Courts now demand exactly this test: lawful authority, a prima facie nexus, the transaction and amount identified, and a proportionate hold [12].

SOLUTION

LienLine turns a complaint and the bank ledgers into a provably proportionate hold, per account.

Read

The police notice and NCRP complaint, cross-checked against real UTRs

Trace

Every hop across banks under four legal tracing rules

Decide

Lien the traceable amount, freeze fully only with recorded reasons, release the rest

Act

Letters, grievance clocks and an audit trail; a nodal officer approves

FOR BANKS

Fewer writ petitions, a defensible record for every hold, the SC-ordered debit-hold SOP [3] met in software.

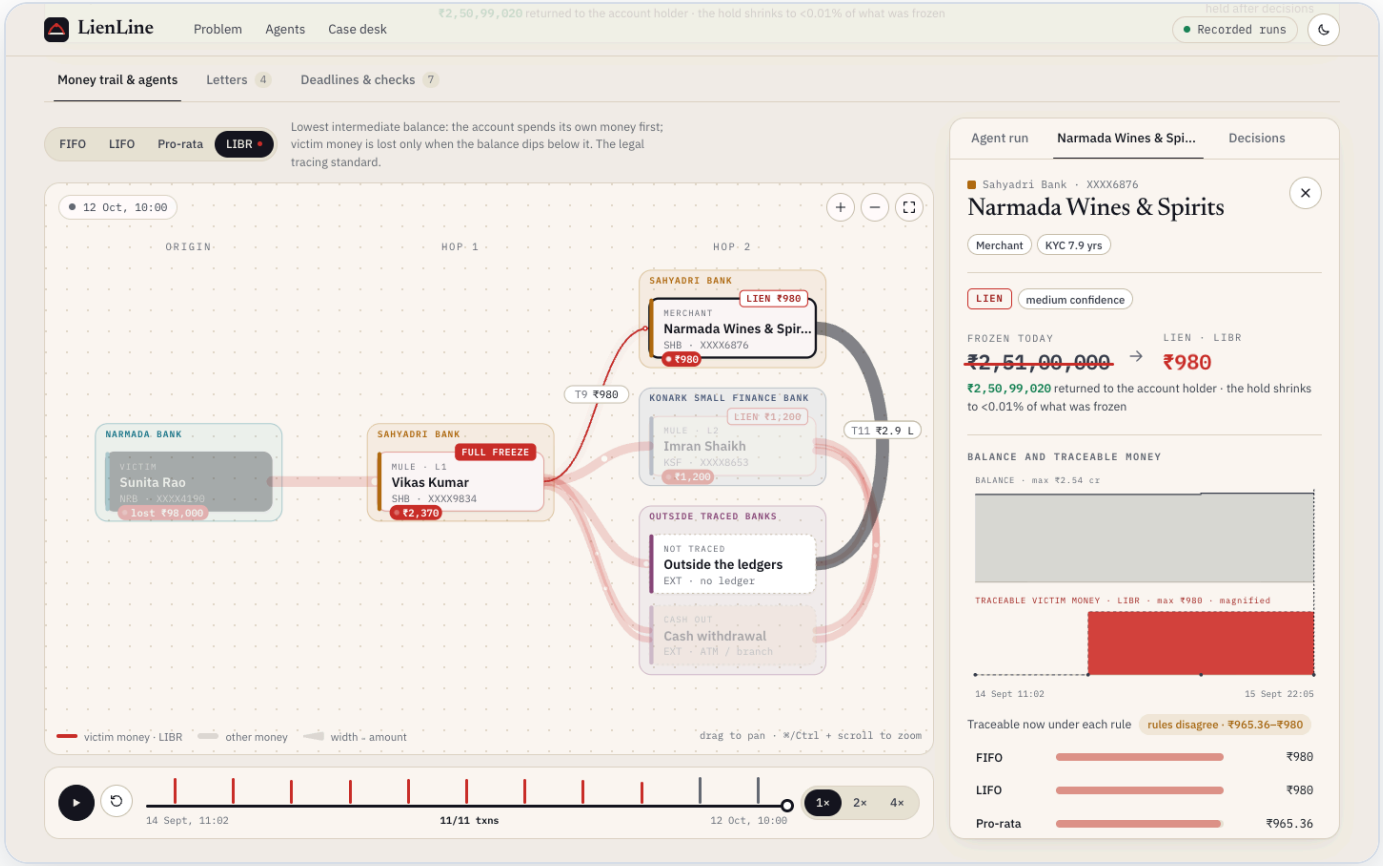
FOR VICTIMS

Every traceable rupee stays on hold; nothing recoverable is released.

FOR MERCHANTS

A ₹980 payment no longer locks a ₹2.51 crore business account.

It already runs: replay a case and watch the money, the agents and the hold change.



- **Money-flow graph** animates each hop; red is the traceable share.
- **Rule switcher** shows FIFO, LIFO, pro-rata and LIBR side by side.
- **Agent console** replays the run: claims, rebuttals, checks.
- **Officer approves** each hold; letters are generated and number-checked.

PROPOSED TECHNOLOGICAL ARCHITECTURE

model · code · person

Nine agents: models read and argue, code computes and verifies, a person decides.

Intake
reads notice + complaint; code checks every UTR

Tracer
four tracing rules over all ledgers

Pattern analyst
mule signals computed in code, explained by model

Two advocates
victim vs account holder argue each hold, must cite

Decider
policy: lien LIBR amount; full freeze needs recorded reasons

Clock keeper
SOP and court deadlines, each with a citation

Drafter
lien, release and grievance letters

Verifier
independent replay; every number re-derived

Nodal officer
approves, edits or rejects each hold

DATA IN

CFCFRMS notice, NCRP complaint, bank ledgers / CBS extracts, Suspect Registry hit

STACK

Python · FastAPI · Postgres · React · any OpenAI-compatible model, or fully on-prem with a local model

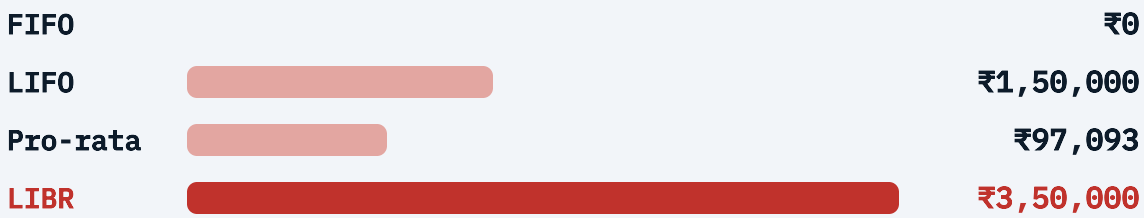
RULES

The model never sets an amount · uncited claims are rejected · a person approves every hold

ARCHITECTURE · HOW THE AMOUNT IS COMPUTED AND TRUSTED

Four legal tracing rules give four answers; LienLine shows all four and holds the defensible one.

MALWA PHARMA DISTRIBUTORS: TRACEABLE FRAUD MONEY BY RULE



Real output of our tracing engine on case 3 (synthetic ledgers). LIBR, the lowest intermediate balance rule, never holds more than the account kept since the fraud money arrived.

- **Hop by hop:** an outflow carries the tainted share given by the rule to the next account.
- **Independent verifier:** replays ledgers; held + cashed out + untraced must equal the amount lost.
- **Advocates must cite:** a claim citing a transaction that does not exist is struck out.
- **Letters:** every number in every letter is re-derived before an officer sees it.

PROBLEM STATEMENT · WHY NOW

In 2026 courts, the Supreme Court and RBI all moved at once; the tool they assume does not exist.



Dated court, regulator and press events on account freezes we found per month, Sep 2024 – Oct 2026 (30 events)

2024-09 Madras HC: freeze only the fraud quantum [24]

2025-12 One scam freezes 36,000+ accounts [6]

2026-07 MP HC statewide rules (₹2.51 crore over ₹980) [1]

2026-09 RBI draft: banks' own 60-day holds [10]

2025-02 Delhi HC: ₹93.5 crore frozen over ₹200 [7]

2026-01 MHA/I4C SOP: grievance clocks [9]

2026-08 SC orders an RBI debit-hold SOP; Rajasthan HC 105 petitions [3]

GLOBAL PRODUCT BENCHMARKING · MULTI-COUNTRY RESEARCH

Six countries fight scams; none sizes the hold on a receiving account from the money trail.

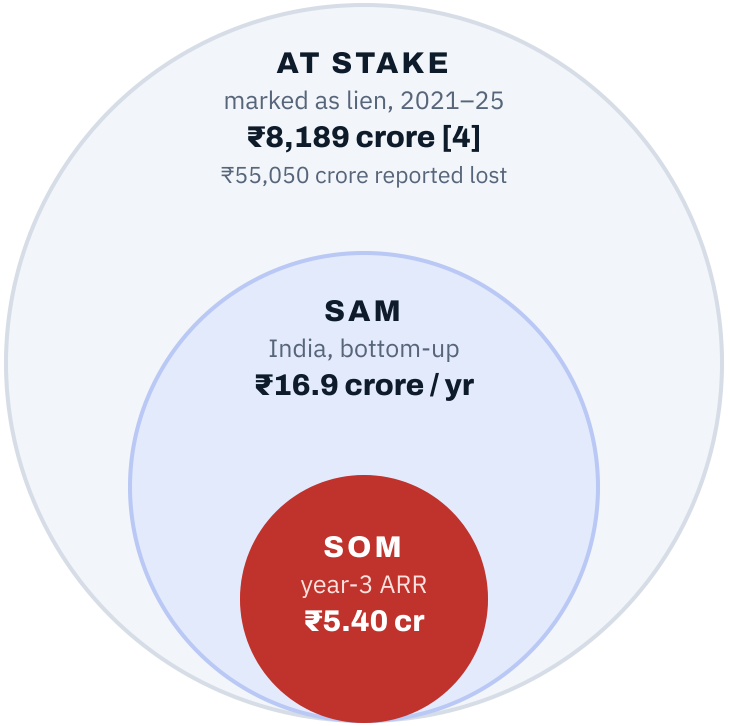
COUNTRY	SYSTEM	WHAT IT DOES	GAP LIENLINE FILLS
India	**CFCFRMS + MuleHunter.AI**	Holds on notice; mule detection in 31 banks [19]	⊗ No per-account amount from the trail
UK	**PSR APP reimbursement**	88% of losses refunded, £316m in 18 months [14]	⊗ Victim refunds, not receiving-side holds
Singapore	**Protection from Scams Act**	18 restriction orders, S$97m recovered, H1 2026 [15]	⊗ Person-level orders, not amount-proportionate
Australia	**Scam-Safe Accord · AFCX FRX**	Industry fraud exchange; A$100m payee check [16]	⊗ Receiving bank decides how much to freeze
USA	**FinCEN 314(b) · Zelle**	Voluntary info sharing; 47% reimbursed [17]	⊗ No tracing-to-hold computation
EU	**Verification of Payee**	Mandatory in euro area since Oct 2025 [18]	⊗ Pre-payment only; nothing after funds land

Prior-art sweep (71 searches): closest is CashGuard-AI, an SIH 2026 fund-block queue with no tracing or lien sizing [20]; MuleHunter.AI detects mules but does not size holds [19].

MARKET SIZE

Projection: our assumptions

A ₹17 crore-a-year serviceable market in India, protecting ₹8,189 crore already under lien.



Circles not to scale

SAM, BUILT BOTTOM-UP

49 commercial, small finance and payments banks (approx.) 49 × ₹24 lakh a year [23]	₹11.8 cr
28 regional rural banks (approx.) 28 × ₹3 lakh a year [23]	₹0.8 cr
36 States/UTs running cyber cells under the SC order 36 × ₹12 lakh a year [3]	₹4.3 cr

Year 3: 14 banks, 20 RRBs, 12 states
₹5.40 crore ARR, 32% of SAM (projection)

Sources: [3] [4] [23]

Projection: our assumptions

Annual licences per bank and per state; break-even in year 2, 43% margin in year 3.

Bank licence
₹24 lakh / yr
Commercial, small finance and payments banks; on-prem option

RRB / co-op
₹3 lakh / yr
Sold through sponsor banks and federations

State cyber cell
₹12 lakh / yr
Lien sizing before notices go out

CAC
₹6 / 0.5 / 2 lakh
Bank / RRB / state; payback < 4 months

₹ LAKH	REVENUE	TOTAL COST	PROFIT
Year 1	24.0	75.5	-51.5
Year 2	240.0	190.0	+50.0
Year 3	540.0	310.0	+230.0

- **Q1–Q2:** two paid bank pilots (half price) and two state cyber cells.
- **Channel:** I4C's Cyber Fraud Mitigation Centre seats banks and intermediaries together.
- **Wedge:** courts and the SC order make proportionate holds mandatory [1] [2] [3].

Sources: [1] [2] [3]

Projection: our assumptions

About ₹20 lakh a year to run, ₹2.4 of model cost per case; year-1 R&D ₹20 lakh.

ITEM	BASIS	₹ / MONTH
Application servers (x2)	Azure D4s_v5, Central India, $0.202/h [22]	₹25,953
Managed Postgres (ledger + audit)	Azure PostgreSQL 4 vCore, $0.502/h [22]	₹32,248
Model inference	25,000 cases x 9 calls, Gemini 3.1 Flash-Lite [21]	₹59,400
Monitoring, backups, logs	*estimate*	₹20,000
Security audit (CERT-In empanelled)	*Rs 3 lakh a year, estimate*	₹25,000
Total · ₹19.5 lakh a year		**₹1,62,601**

YEAR-1 R&D	₹ LAKH
Synthetic multi-bank ledger benchmark + legal-rule encoding (lawyer review)	6.0
CBS / FRMS / CFCFRMS connectors in a bank sandbox	6.0
Model evaluation and red-teaming of the advocate agents	3.0
Hindi and regional-language notices	2.0
On-prem packaging for air-gapped banks	3.0
Total year 1	**20.0**

At 25,000 cases a month; USD at ₹88. Banks that need air-gapped deployment run a local model on their own hardware, removing the inference line.

Sources: [21] [22]

VALIDATION

Across four scenarios, frozen money fell from ₹2.75 crore to ₹6.7 lakh while every traceable rupee stayed on hold.

SCENARIO (SYNTHETIC MULTI-BANK LEDGERS)	LOST	HELD TODAY	LIENLINE HOLD	VERIFIER
₹980 dispute, ₹2.51 crore frozen	₹98,000	₹2.51 crore	₹4,550	verified
₹18.4 lakh 'digital arrest', 8 accounts frozen	₹18.4 lakh	₹6.1 lakh	₹2.9 lakh	verified
One distributor, four tracing rules, four answers	₹6.0 lakh	₹16.6 lakh	₹3.6 lakh	verified
A mule on 7 complaints: full freeze, justified	₹3.2 lakh	₹1.1 lakh	₹21,300	verified

INNOCENT MONEY RELEASED

98%

of what whole-account freezes held

VICTIM PROTECTION

100%

of traceable money still on hold (conservation check)

EVERY NUMBER CHECKED

4 / 4

runs pass the independent verifier

Sources: [1] [6] [7] [8]

RISKS · ROADMAP · THE ASK

Pilot with one bank and one state cyber cell, and measure holds released and petitions avoided.

RISKS AND HOW WE HANDLE THEM

- **Legal rule unsettled:** show all four rules; policy rule is configurable per bank's counsel.
- **Data sensitivity:** on-prem with a local model; no data leaves the bank.
- **RBI draft may entrench whole-account holds [10]:** LienLine documents the reasons RBI's draft requires.

ROADMAP

- **0–3 months:** bank sandbox, CFCFRMS notice format, masked historic cases.
- **3–9 months:** two paid pilots; Hindi and regional notices.
- **9–18 months:** RRBs via sponsor banks; state cyber cells before notices are sent.

The ask. One introduction to a bank's fraud-operations head and one state cyber cell; Malansoft's ThreatSentry can carry LienLine's case file and chain of custody.

VALIDATED REFERENCES

25 sources behind every figure, each opened and checked in October 2026

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